

ON 2-ADIC DEFECT RIGIDITY FOR THE DICKSON–PILLAI CONDITION: EXACT VALUATIONS, DISCRETE CARRY DRIFT, AND THE ARCHIMEDEAN BARRIER

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ABSTRACT. The Dickson–Pillai condition $r_k + q_k \leq 2^k$ (where $3^k = q_k 2^k + r_k$) governs the universal formula $g(k) = 2^k + \lfloor (3/2)^k \rfloor - 2$ for Waring’s problem across all $k \geq 6$. In this paper, we investigate the 2-adic algebraic structure of the shifted Diophantine defects $D_k := 2^k - r_k = (q_k + 1)2^k - 3^k$ in the ring $\mathbb{Z}$.

We establish exact algebraic valuation identities governing shifted defects: (1) For all even $k \geq 6$, the Lifting the Exponent Lemma (LTE) combined with the ultrametric inequality proves $v_2(D_k + 1) = v_2(k) + 2$; (2) For odd $k \equiv 3 \pmod{4}$ ($k \geq 5$), the defect is rigidly locked to $D_k \equiv 5 \pmod{16}$, yielding $v_2(D_k + 3) = 3$; (3) For odd $k \equiv 1 \pmod{4}$, $v_2(D_k + 3) = v_2(k - 1) + 2$. As an immediate corollary, $D_k \bmod 8 \in \{5, 7\}$ for all $k \geq 3$, which unconditionally proves that small defects $D_k \in \{1, 3\}$ have zero solutions.

Furthermore, we analyze the discrete carry drift recurrence $3D_k - D_{k+1} = C_k 2^k$ and prove that the multiplier $C_k = 3q_k - 2q_{k+1} + 1$ takes values in $\{-1, 0, 1, 2\}$, disproving the naive conjecture that $C_k \in \{-1, 0, 1\}$ (witnessed by $C_4 = C_7 = 2$). Finally, we elucidate the fundamental *Archimedean vs. 2-Adic Barrier*: because the 2-adic modulus is bounded by $2^{v_2(k)+3} \leq 8k$ while the failure threshold $q_k \asymp (1.5)^k$ grows exponentially, there exist $\asymp (1.5)^k / (8k)$ candidate integers in the danger zone $[1, q_k]$ satisfying the exact 2-adic conditions. Consequently, 2-adic rigidity alone cannot establish the Archimedean inequality $D_k \geq q_k$. All division relations, parity valuations, and range verifications are formally formalized in Lean 4 with zero external axioms.

1. INTRODUCTION AND PROBLEM FORMULATION

For each integer $k \geq 1$, let $g(k)$ denote the minimal number of non-negative k -th powers needed to represent every positive integer. In 1936, Dickson [1] and Pillai [4] proved that for all $k \geq 6$:

$$g(k) = 2^k + \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor - 2, \quad (1.1)$$

provided that the remainder $r_k = 3^k \bmod 2^k$ and quotient $q_k = \lfloor (3/2)^k \rfloor$ satisfy the *Dickson–Pillai condition*:

$$r_k + q_k \leq 2^k, \quad \text{where } 3^k = q_k 2^k + r_k \quad (0 \leq r_k < 2^k). \quad (1.2)$$

Defining the Diophantine *defect* D_k by

$$D_k := 2^k - r_k = (q_k + 1)2^k - 3^k, \quad (1.3)$$

the condition (1.2) is strictly equivalent to:

$$D_k \geq q_k. \quad (1.4)$$

A failure of the condition would correspond to an exponent k where $1 \leq D_k < q_k$. Although Kubina and Wunderlich [2] computationally verified that no exceptions exist for $k \leq 471,600,000$, and Mahler [3] proved that only finitely many exceptions can exist via Ridout’s p -adic Thue–Siegel–Roth theorem, an effective proof that *zero* exceptions exist remains one of the notorious open problems in Diophantine number theory.

In this work, we analyze the 2-adic valuation rigidity of D_k and the discrete carry recurrence governing transitions $D_k \mapsto D_{k+1}$.

2. THE 2-ADIC DEFECT RIGIDITY THEOREMS

Let $v_2(x)$ denote the standard 2-adic valuation on $\mathbb{Z}$, with the ultrametric property:

$$v_2(A - B) = \min(v_2(A), v_2(B)) \quad \text{whenever } v_2(A) \neq v_2(B).$$

2.1. Even Index Rigidity ($k = 2m$).

Theorem 2.1 (Even Index 2-Adic Rigidity). *For every even integer $k = 2m$ with $k \geq 6$:*

$$v_2(D_k + 1) = v_2(k) + 2. \quad (2.1)$$

Proof. From the defect identity (1.3), we add 1 to both sides:

$$D_k + 1 = (q_k + 1)2^k - (3^k - 1) = (q_k + 1)2^k - (9^m - 1). \quad (2.2)$$

We factor $9^m - 1 = (8 + 1)^m - 1 = 8V_m$, where $V_m := \sum_{\ell=0}^{m-1} 9^\ell$. By the classical Lifting the Exponent Lemma (LTE) for $p = 2$:

$$v_2(9^m - 1) = v_2(9 - 1) + v_2(m) = 3 + v_2(m) = 3 + (v_2(k) - 1) = v_2(k) + 2. \quad (2.3)$$

Now, we inspect the valuation of the first term $A = (q_k + 1)2^k$:

$$v_2(A) = k + v_2(q_k + 1) \geq k.$$

For all even $k \geq 6$, we have $k > v_2(k) + 2$. Indeed:

- If $k = 6$: $v_2(6) + 2 = 1 + 2 = 3 < 6$.
- If $k = 8$: $v_2(8) + 2 = 3 + 2 = 5 < 8$.
- In general, for $k \geq 6$, $2^{k/2} > k$, so $v_2(k) \leq \log_2 k < k - 2$.

Applying the non-Archimedean ultrametric property to $D_k + 1 = A - (9^m - 1)$:

$$v_2(D_k + 1) = \min(v_2(A), v_2(9^m - 1)) = v_2(9^m - 1) = v_2(k) + 2. \quad \square$$

2.2. Odd Index Rigidity ($k \equiv 3 \pmod{4}$).

Theorem 2.2 (Odd Branch $k \equiv 3 \pmod{4}$ Rigidity). *For every integer $k \equiv 3 \pmod{4}$ with $k \geq 5$:*

$$v_2(D_k + 3) = 3 \iff D_k \equiv 5 \pmod{16}. \quad (2.4)$$

Proof. Let $k = 2m + 1$ with m odd. Then $3^k = 3 \cdot 9^m = 3 + 3(9^m - 1) = 3 + 24V_m$. Adding 3 to D_k :

$$D_k + 3 = (q_k + 1)2^k - (3^k - 3) = (q_k + 1)2^k - 24V_m.$$

Dividing by 8:

$$\frac{D_k + 3}{8} = (q_k + 1)2^{k-3} - 3V_m.$$

Since m is odd, $V_m = \sum_{\ell=0}^{m-1} 9^\ell \equiv m \cdot 1 \equiv 1 \pmod{2}$, so $3V_m$ is odd. For $k \geq 5$, $k - 3 \geq 2$, hence $(q_k + 1)2^{k-3}$ is strictly even. The difference of an even and an odd integer is odd, so $(D_k + 3)/8$ is odd, which establishes $v_2(D_k + 3) = 3$, or equivalently $D_k + 3 \equiv 8 \pmod{16} \implies D_k \equiv 5 \pmod{16}$. $\square$

2.3. Odd Index Branch ($k \equiv 1 \pmod{4}$).

Theorem 2.3 (Odd Branch $k \equiv 1 \pmod{4}$ Rigidity). *For every integer $k \equiv 1 \pmod{4}$ with $k - v_2(k - 1) \geq 3$:*

$$v_2(D_k + 3) = v_2(k - 1) + 2. \quad (2.5)$$

Proof. When $m = (k - 1)/2$ is even, $v_2(m) = v_2(k - 1) - 1$. Then $v_2(24V_m) = 3 + v_2(m) = v_2(k - 1) + 2$. Since $k - v_2(k - 1) \geq 3 \iff k > v_2(k - 1) + 2$, the term $(q_k + 1)2^k$ has strictly higher valuation, and the ultrametric property yields $v_2(D_k + 3) = v_2(k - 1) + 2$. $\square$

3. SMALL DEFECT EXCLUSIONS AND MODULAR RESTRAINTS

Theorem 3.1 (Residue Exclusion Modulo 8 and Defect Non-Triviality). *For all $k \geq 3$:*

$$D_k \bmod 8 = \begin{cases} 7 & \text{if } k \text{ is even,} \\ 5 & \text{if } k \text{ is odd.} \end{cases} \quad (3.1)$$

Consequently, $D_k \neq 1$ and $D_k \neq 3$ for all $k \geq 3$.

Proof. For $k \geq 3$, $2^k \equiv 0 \pmod{8}$. Therefore,

$$D_k \equiv -3^k \pmod{8}.$$

By elementary modular arithmetic, $3^k \equiv 1 \pmod{8}$ for even k and $3^k \equiv 3 \pmod{8}$ for odd k . Thus:

$$D_k \equiv -1 \equiv 7 \pmod{8} \quad (k \text{ even}), \quad D_k \equiv -3 \equiv 5 \pmod{8} \quad (k \text{ odd}).$$

Because $1 \not\equiv 5, 7 \pmod{8}$ and $3 \not\equiv 5, 7 \pmod{8}$, the defect can never equal 1 or 3. $\square$

Remark 3.2. This theorem proves that the smallest defects $D_k \in \{1, 3\}$ have zero solutions unconditionally. Furthermore, since 3^k is odd and 2^k is even, r_k is odd, which forces $D_k = 2^k - r_k$ to be strictly odd for all $k \geq 1$. Hence, D_k is never even.

4. THE DISCRETE CARRY DRIFT RECURRENCE AND MULTIPLIER SPECTRUM

Theorem 4.1 (Exact Discrete Drift Recurrence). *For all $k \geq 1$:*

$$3D_k - D_{k+1} = C_k 2^k, \quad (4.1)$$

where the carry multiplier C_k is defined by:

$$C_k := 3q_k - 2q_{k+1} + 1. \quad (4.2)$$

Furthermore, the multiplier spectrum satisfies:

$$C_k \in \{-1, 0, 1, 2\}. \quad (4.3)$$

Proof. Using $D_k = (q_k + 1)2^k - 3^k$ and $D_{k+1} = (q_{k+1} + 1)2^{k+1} - 3^{k+1}$:

$$\begin{aligned} 3D_k - D_{k+1} &= 3(q_k + 1)2^k - 3 \cdot 3^k - [(q_{k+1} + 1)2 \cdot 2^k - 3 \cdot 3^k] \\ &= [3(q_k + 1) - 2(q_{k+1} + 1)]2^k \\ &= [3q_k - 2q_{k+1} + 1]2^k = C_k 2^k. \end{aligned}$$

To determine the bounds on C_k , note that $3^{k+1} = 3(q_k 2^k + r_k) = 3q_k 2^k + 3r_k$, while $3^{k+1} = 2q_{k+1} 2^k + r_{k+1}$. Equating the two yields:

$$2q_{k+1} - 3q_k = \frac{3r_k - r_{k+1}}{2^k}.$$

Since $0 \leq r_k < 2^k$ and $0 \leq r_{k+1} < 2^{k+1} = 2 \cdot 2^k$, we have:

$$-2 \cdot 2^k < 3r_k - r_{k+1} < 3 \cdot 2^k.$$

Because $2q_{k+1} - 3q_k \in \mathbb{Z}$, it can only take values in $\{-1, 0, 1, 2\}$. Substituting into $C_k = 1 - (2q_{k+1} - 3q_k)$ gives $C_k \in \{-1, 0, 1, 2\}$. $\square$

Remark 4.2 (Disproof of the Naive $C_k \in \{-1, 0, 1\}$ Conjecture). It is tempting to conjecture that $C_k \in \{-1, 0, 1\}$. However, explicit computation disproves this claim:

- At $k = 4$: $q_4 = 5, q_5 = 7 \implies C_4 = 3(5) - 2(7) + 1 = 15 - 14 + 1 = 2$.
- At $k = 7$: $q_7 = 17, q_8 = 25 \implies C_7 = 3(17) - 2(25) + 1 = 51 - 50 + 1 = 2$.
- At $k = 16$: $q_{16} = 656, q_{17} = 985 \implies C_{16} = 3(656) - 2(985) + 1 = 1968 - 1970 + 1 = -1$.

Thus, C_k genuinely attains the full 4-element spectrum $\{-1, 0, 1, 2\}$.

5. THE FUNDAMENTAL ARCHIMEDEAN VS. 2-ADIC BARRIER

We now address why the 2-adic rigidity theorems, despite being exact and unconditionally true, *cannot prove* the Dickson–Pillai condition $D_k \geq q_k$.

TABLE 1. Comparison of the 2-Adic Framework vs. Dickson–Pillai Archimedean Barrier

Metric / Property	2-Adic Rigidity Framework	Dickson–Pillai Conjecture
Topology	Non-Archimedean $ \cdot _2$ on $\mathbb{Q}_2$	Archimedean $ \cdot _\infty$ on $\mathbb{R}$
Governing Information	Lowest set bit ($v_2(D_k + c)$)	Highest set bit (Magnitude $ D_k $)
Modulus / Resolution	$2^{v_2(k)+3} \leq 8k$ ($O(k)$ bits)	Exponential barrier $q_k \approx (1.5)^k$
Excluded Set	$D_k \in \{1, 3\}$	All integers in $[1, q_k)$
Leakage Window	$\asymp (1.5)^k / (8k)$ candidates	Requires 0 candidates

Theorem 5.1 (Archimedean Candidate Leakage). *For even k , the 2-adic condition $v_2(D_k + 1) = v_2(k) + 2$ is equivalent to a single arithmetic progression:*

$$D_k \equiv 2^{v_2(k)+2} - 1 \pmod{2^{v_2(k)+3}}. \quad (5.1)$$

For all $k \geq 10$, the interval $[1, q_k)$ contains strictly positive numbers of integers satisfying this exact congruence:

$$N_{\text{cand}}(k) := \#\left\{d \in [1, q_k) : d \equiv 2^{v_2(k)+2} - 1 \pmod{2^{v_2(k)+3}}\right\} \asymp \frac{(1.5)^k}{8k} \rightarrow \infty. \quad (5.2)$$

Proof. The modulus $M_k = 2^{v_2(k)+3}$ satisfies $M_k \leq 8k$. The interval length is $q_k = \lfloor (3/2)^k \rfloor$. The number of integers in $[1, q_k)$ congruent to a fixed residue modulo M_k is $\lfloor (q_k - 1)/M_k \rfloor$ or $\lceil (q_k - 1)/M_k \rceil$. Since $(1.5)^k$ grows exponentially while $8k$ grows linearly:

$$\frac{q_k}{M_k} \geq \frac{(1.5)^k - 1}{8k} \gg 1 \quad (\forall k \geq 10).$$

For instance, at $k = 10$, $q_{10} = 57$, while $M_{10} = 16$. The residue class is $d \equiv 7 \pmod{16}$. The integers $d \in \{7, 23, 39\}$ all satisfy $v_2(d + 1) = 3 = v_2(10) + 2$ and all lie strictly inside $[1, q_{10})$. At $k = 100$, $q_{100} \approx 4.06 \times 10^{17}$ and $M_{100} = 32$, yielding over 1.2×10^{16} candidate failure values satisfying the exact 2-adic rigidity condition. $\square$

Corollary 5.2. *No argument operating purely inside the non-Archimedean metric $\mathbb{Q}_2$ (such as 2-adic valuations or modular carry recurrences) can prove $D_k \geq q_k$.*

6. FORMALIZATION IN LEAN 4

The algebraic foundations, modular exclusions, multiplier spectrum, and Archimedean gap theorems have been formalized in Lean 4 in the companion file:

`formalization/TwoAdicDefectRigidity.lean`

The formalization is verified with zero unproven axioms (relying only on standard Lean core `propext`, `Quot.sound`, and `Classical.choice`).

REFERENCES

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